

## Multiple Choice (10 marks)

1. Let `x = 8`. What is `x >> 1` in Python 3?
  - (a) 3
  - (b) 4
  - (c) 2
  - (d) 8
2. Which of the following best explains what happens when a new device is connected to the Internet?
  - (a) A device driver is assigned to the device.
  - (b) An Internet Protocol (IP) address is assigned to the device.
  - (c) A packet number is assigned to the device.
  - (d) A Web site is assigned to the device.
3. What is the output of `"abc"[-1]` in Python 3?
  - (a) Error
  - (b) a
  - (c) b
  - (d) c
4. In Python 3, which of the following function removes all leading and trailing whitespace in string?
  - (a) `replace(old, new [, max])`
  - (b) `strip([chars])`
  - (c) `swapcase()`
  - (d) `title()`
5. Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-ofservice (D o S) attack?
  - (a) The goal of the attack
  - (b) The number of computers being attacked
  - (c) The number of computers launching the attack
  - (d) The time period in which the attack occurs

## True / False (10 marks)

*Answer True or False*

6. True or False: In Python 3, the expression `['Hi!'] * 4` results in an error due to incorrect syntax.
7. True or False:  $O(1)$  represents the smallest asymptotic complexity, indicating constant time regardless of input size.
8. True or False:  $O(n^2)$  grows faster than  $O(n)$  as the input size increases, making it the largest asymptotic notation among the options.
9. True or False: In Python 3, floor division is performed using the single forward slash operator `'/'`.
10. True or False: A function that sorts values in a column or row can help identify unusually high or low data entry errors.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss how the Python function `'set()'` processes a list with duplicate elements, using the list `l = [1, 2, 2, 3, 4]` as an example. What output would you expect and why?
12. Discuss how the observation that computer processing speeds double approximately every two years can influence technology companies' research and development strategies.

*End of Paper*